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## Chemistry Paper 3A WASSCE (SC), 2023

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### Question 3A

a. State what would be observed if few drops of NaOH solution is added to 2 cm<sup>3</sup> of solutions of each of the following salts:

- $\text{Pb}(\text{NO}_3)_2$ ;
- $\text{Fe}_2(\text{SO}_4)_2$ .

[2 marks]

b. Describe briefly how the melting point of benzoic acid could be determined in the laboratory.

[6 marks]

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### Observation

This question was based on the knowledge of Test of Practical, and was poorly answered by the candidates.

In part (a), majority of the candidates could not state what would be observed if few drops of NaOH solution is added to 2 cm<sup>3</sup> of solutions of  $\text{Pb}(\text{NO}_3)_2$  and  $\text{Fe}_2(\text{SO}_4)_2$ .

In part (b), majority of the candidates could not describe how the melting point of benzoic acid could be determined in the laboratory.

The expected answers include:

- (i) White (chalky) precipitate is formed
- (ii) (Reddish) brown (gelatinous) precipitate is formed

(b) A small quantity of benzoic acid crystals is placed into a capillary tube sealed at one end the filled capillary tube is tied to a thermometer.

The set-up is clamped and lowered into a beaker half filled with liquid paraffin. The beaker is heated gently over Bunsen flame and the rise in temperature observed

The temperature at which the crystal begins to liquefy (ti) and also the temperature

at which it is completely liquefied ( $t_2$ ) are noted. The melting point is the range ( $t_1 - t_2$ ) °C.

#### Alternatives

- Place a small quantity of the powdered benzoic acid into a capillary tube sealed at one end. Tie it to the thermometer such that the bulb of the thermometer is completely immersed in the beaker, that is half filled with liquid paraffin and placed on a tripod stand with wire gauze.
- Heat the content of the beaker gently with constant stirring of the liquid paraffin
- Note the temperature at which the benzoic acid melts.
- Repeat the process using a fresh/ new benzoic acid until the readings agree within 2.0°C.

For candidates using melting point apparatus:

Grind the benzoic acid into powder

Fill the capillary tube with the powder and insert into the melting point apparatus

Heat the set up gradually and note the temperature at which the benzoic acid just melts

Read/record the temperature of the thermometer

Repeat the process using fresh powder until two melting points agree within 2.0°C

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